



Semester One Examination, 2025

Question/Answer booklet

MATHEMATICS METHODS UNIT 1

Section One: Calculator-free

SOLUTIONS

WA student number: In figures

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In words

Your name

Time allowed for this section

Reading time before commencing work: five minutes

Working time: fifty minutes

Number of additional
answer booklets used
(if applicable):

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Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet

Formula sheet

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	52	35
Section Two: Calculator-assumed	12	12	100	98	65
Total					100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
3. You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
4. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
5. It is recommended that you do not use pencil, except in diagrams.
6. Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section One: Calculator-free

35% (52 Marks)

This section has **seven** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 50 minutes.

Question 1

(6 marks)

- (a) Determine the equation of the straight line through the point $(2, -7)$ that is perpendicular to the line with equation $y = -\frac{1}{4}x + 1$. (2 marks)

Solution
Slope: $m = -1 \div \left(-\frac{1}{4}\right) = 4$ Equation: $y + 7 = 4(x - 2) \text{ or } y = 4x - 15$
Specific behaviours
✓ indicates correct slope ✓ correct equation

- (b) Function g is defined by $g(t) = 6t^7 - 3t^5 - 11t^4 - t^3 + 5t^2 - 2$. State

- (i) the degree of g . (1 mark)

Solution
Degree is 7.
Specific behaviours
✓ correct degree

- (ii) the coefficient of the t^5 term. (1 mark)

Solution
Coefficient is -3 .
Specific behaviours
✓ correct coefficient

- (c) Solve the equation $(x - 7)^2 = 3$. (2 marks)

Solution
$x - 7 = \pm\sqrt{3}$ $x = 7 \pm \sqrt{3}$
Specific behaviours
✓ indicates valid steps to obtain solutions ✓ two correct solutions

AO ok

Question 2

(8 marks)

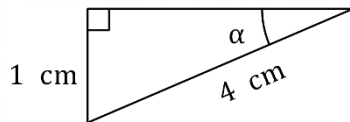
- (a) Express the angle
- 9°
- in radian measure.

(1 mark)

Solution
$\frac{9^\circ \times \pi}{180^\circ} = \frac{\pi}{20}$ radians
Specific behaviours
✓ correct <u>simplified</u> radian measure

- (b) Determine the value of
- $\tan \alpha$
- in the right triangle shown below.

(2 marks)



Solution
Length of adjacent side is $\sqrt{4^2 - 1^2} = \sqrt{15}$. $\tan \alpha = \frac{1}{\sqrt{15}} = \frac{\sqrt{15}}{15}$
Specific behaviours
✓ length of adjacent side ✓ correct value

- (c) Solve the equation
- $\sin(2x) \times \cos\left(2x - \frac{\pi}{2}\right) = 1$
- for
- $0 \leq x \leq \pi$
- .

(5 marks)

Solution	
$\sin(2x) \left(\cos(2x) \cos\left(\frac{\pi}{2}\right) + \sin(2x) \sin\left(\frac{\pi}{2}\right) \right) = 1$ $\sin(2x) \sin(2x) = 1$ $\sin^2(2x) = 1$ $\sin(2x) = \pm 1$ $2x = \frac{\pi}{2}, \frac{3\pi}{2} \text{ for } 0 \leq 2x \leq 2\pi$ $x = \frac{\pi}{4}, \frac{3\pi}{4}$	$\sin(2x) \left(\cos\left(2\left(x - \frac{\pi}{4}\right)\right) \right) = 1$ $\sin(2x) \sin(2x) = 1$ $\sin^2(2x) = 1$ $\sin(2x) = \pm 1$ $2x = \frac{\pi}{2}, \frac{3\pi}{2} \text{ for } 0 \leq 2x \leq 2\pi$ $x = \frac{\pi}{4}, \frac{3\pi}{4}$
Specific behaviours	
✓ uses angle sum identity on $\cos\left(2x - \frac{\pi}{2}\right)$ correctly	✓ uses the idea of transformations $\sin(x) = \cos\left(x - \frac{\pi}{2}\right)$ correctly
✓ solves $\cos\left(2x - \frac{\pi}{2}\right) = \sin(2x)$ correctly ✓ rearranges for $\sin(2x)$ ✓ obtains at least one correct value for $2x$ ✓ correct solutions	

Question 3

(7 marks)

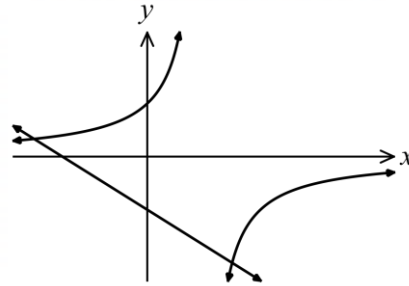
The graphs of the hyperbola $y = \frac{c}{2-x}$

and the straight line $x + y + 3 = 0$ are

shown at right, where c is a constant.

The hyperbola cuts the vertical axis

when $y = 3$.



- (a) Determine the slope of the straight line.

(1 mark)

Solution
$y = -x - 3 \Rightarrow$ slope is -1
Specific behaviours
✓ correct slope

- (b) Determine the value of c .

(1 mark)

Solution
$3 = \frac{c}{2-0} \Rightarrow c = 6$
Specific behaviours
✓ correct value

- (c) State the equation of each asymptote of the hyperbola.

(2 marks)

Solution
Equations are $y = 0$ and $x = 2$.
Specific behaviours
✓ horizontal asymptote
✓ vertical asymptote

- (d) Determine the x -coordinate of each point of intersection of the hyperbola and the straight line.
(3 marks)

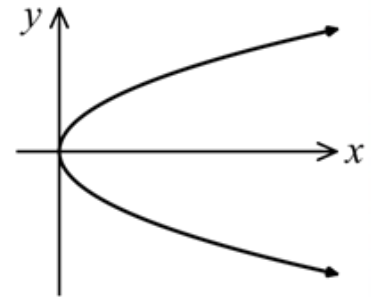
Solution
$\frac{6}{2-x} = -x - 3$ $6 = (2-x)(-x-3)$ $6 = -2x - 6 + x^2 + 3x$ $x^2 + x - 12 = 0$ $(x+4)(x-3) = 0$ $x = -4, x = 3$
Specific behaviours
✓ forms equation and cross multiplies ✓ expands and equates to 0 ✓ factorises and states both x -coordinates

Question 4**(7 marks)**

The relation $y^2 = x$ passes through the points $(a, 4)$ and $(9, b)$.

- (a) State all possible values for a and b . (2 marks)

Solution		
$x = a,$	$y = 4 \Rightarrow 4^2 = a,$	$a = 16$
$x = 9,$	$y = b \Rightarrow b^2 = 9,$	$b = \pm 3$
Specific behaviours		
✓ correct value for a		
✓ correct values for b		



- (b) Write down the name given to the shape of the graph of $y^2 = x$. (1 mark)

Solution
The shape is parabolic.
Specific behaviours
✓ correct name

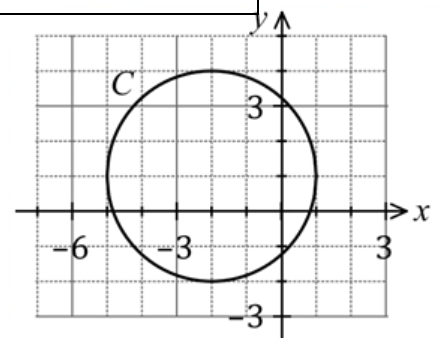
- (c) State, with reasoning, whether the relation $y^2 = x$ is a function or not. (1 mark)

Solution
The relation cannot be expressed as a function since any vertical line of the form $x = c$ ($c > 0$) will cut the curve more than once.
Specific behaviours
✓ states relation cannot be expressed as a function + adequate reasoning
- Sketches vertical line that passes through relation multiple times
- States any vertical line $x = c$ ($c > 0$) will pass through multiple times

The graph of circle C is drawn on a cartesian plane.

- (d) State the radius of circle C . (1 mark)

Solution
$r = 3$
Specific behaviours
✓ correct radius



- (e) Determine the equation of circle C . (2 marks)

Solution
$(x + 2)^2 + (y - 1)^2 = 9$
Specific behaviours
✓ uses correct form of equation
✓ correct equation

Question 5

(10 marks)

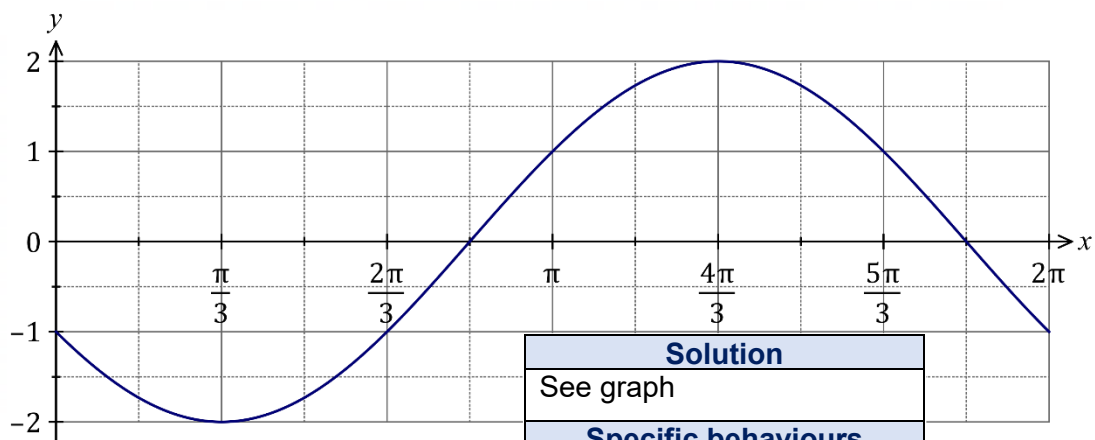
- (a) State the amplitude and period of the function $f(t) = 2 - 5 \sin(4t)$, where $0 \leq t \leq 20\pi$.

(2 marks)

Solution
Amplitude is 5.
Period is $2\pi \div 4 = \frac{\pi}{2}$.
Specific behaviours
✓ correct amplitude
✓ correct period

- (b) Sketch the graph of $y = -2 \cos\left(x - \frac{\pi}{3}\right)$ on the axes below for $0 \leq x \leq 2\pi$.

(4 marks)



Solution
See graph
Specific behaviours
✓ roots
✓ turning points
✓ neat sinusoidal curve
✓ starts and ends at $y = -1$

- (c) Evaluate $\sin\left(\frac{\pi}{12}\right)$.

(4 marks)

Solution
$\sin\left(\frac{\pi}{12}\right) = \sin\left(\frac{\pi}{3} - \frac{\pi}{4}\right)$ $\sin\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = \sin\left(\frac{\pi}{3}\right) \cos\left(\frac{\pi}{4}\right) - \cos\left(\frac{\pi}{3}\right) \sin\left(\frac{\pi}{4}\right)$ $= \frac{\sqrt{3}}{2} \times \frac{\sqrt{2}}{2} - \frac{1}{2} \times \frac{\sqrt{2}}{2}$ $= \frac{\sqrt{6} - \sqrt{2}}{4}$
Specific behaviours
✓ splits $\frac{\pi}{12}$ into $\frac{\pi}{3} - \frac{\pi}{4}$ ✓ indicates use of correct identity ✓ correctly evaluates at least three terms ✓ correct evaluation

Question 6**(7 marks)**

The events A and B are such that $P(A) = 0.4$ and $P(B) = 0.5$.

- (a) Determine $P(A \cup B)$ when $P(A \cap B) = 0.18$.

(2 marks)

Solution
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$ $= 0.4 + 0.5 - 0.18$ $= 0.72$
Specific behaviours
✓ indicates correct use of probability rule ✓ correct probability

AO ok

- (b) Determine $P(A | \bar{B})$ when $P(\bar{A} \cup B) = 0.92$.

(2 marks)

Solution
$P(A \bar{B}) = \frac{P(A \cap \bar{B})}{P(\bar{B})} = \frac{1 - P(\bar{A} \cup B)}{1 - P(B)}$ $= \frac{1 - 0.92}{1 - 0.5} = \frac{0.08}{0.5} = \frac{4}{25} = 0.16$
Specific behaviours
✓ indicates correct denominator for conditional probability ✓ correct probability

- (c) State, with justification, whether A and B are independent events when $P(A \cap \bar{B}) = 0.15$.

(3 marks)

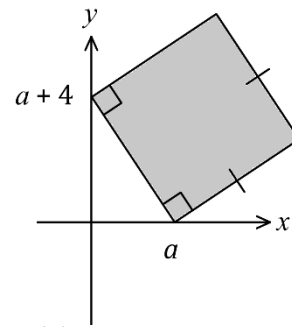
Solution
$P(A \cap \bar{B}) = 0.15 \Rightarrow P(A \cap B) = 0.4 - 0.15 = 0.25$ $P(A) \times P(B) = 0.4 \times 0.5 = 0.2$ Events are not independent since it can be seen that $P(A) \times P(B) \neq P(A \cap B)$.
Specific behaviours
✓ obtains correct $P(A \cap B)$ ✓ obtains correct $P(A) \times P(B)$ ✓ states not independent, with justification

Question 7

(7 marks)

A computer screensaver shows a shaded square that has vertices at $(0, a + 4)$ and $(a, 0)$.

Nick is an animator who is bored of staring at a static image. So, Nick creates a computer program that animates the square by changing the value of a , so that $-6 \leq a \leq 6$. The image moves and resizes but remains a square.



- (a) Show that the area of the square is given by $A = 2a^2 + 8a + 16$. (2 marks)

Solution
Length of square is $l = \sqrt{(a)^2 + (a + 4)^2}$ and area of square is $A = l^2 = a^2 + (a + 4)^2$ $= a^2 + a^2 + 8a + 16$ $= 2a^2 + 8a + 16$
Specific behaviours
✓ correct expression for length of square ✓ shows expansion of both terms before simplifying

- (b) Determine the area of the square when $a = -6$. (1 mark)

Solution
$A = 2a^2 + 8a + 16$ $A(-6) = 2(-6)^2 + 8(-6) + 16$ $= 40 \text{ units}^2$
Specific behaviours
✓ correct area

- (c) Determine the coordinates of the turning point of the graph of $A = 2a^2 + 8a + 16$. (2 marks)

Solution
$A = 2(a^2 + 4a + 8)$ $= 2((a + 2)^2 + 4)$ $= 2(a + 2)^2 + 8$ Turning point is at $(-2, 8)$.
Specific behaviours
✓ indicates any appropriate method ✓ correct coordinates

- (d) Determine the minimum and maximum area of the square during the animation. (2 marks)

Solution
Minimum at turning point, $A_{MIN} = 8$, and maximum at one of domain bounds: $A(-6) = 40, \quad A(6) = 6^2 + 10^2 = 136.$ Minimum is 8 and maximum is 136 square units.
Specific behaviours
✓ correct minimum ✓ correct maximum

Supplementary page

Question number: _____

Supplementary page

Question number: _____

